(3 Shifts) Results

October 13, 2026

1 Experimental Setup

- **CPU:** 11th Gen Intel® Core™ i7-1165G7 @ 2.80 GHz × 8
- **Memory and OS:** [15Gi], [22.04.5 LTS (Jammy Jellyfish)]
- **Software:** [Python 3.10.12], PuLP [3.3.0], CBC [2.10.3]

2 Optimizers

Solvers (ILP) : CBC

-> Configuration for Puzzle Heuristic : Gap limit = 2%, Threads = 4
-> Reference solution : The Exact ILP was solved with a gap limit of 1% and 4 Threads. Throughout this report, its solution (488 ideal shortages) is taken as the optimal reference, and the gaps of the heuristics are measured against it.

3 Restrictions Covered

- Full coverage of minimum staffing requirements per shift and team (*soft constraint*);
- At most one shift assignment per employee per day (*Hard Constraint*);
- At most 5 consecutive working days per employee (*Hard Constraint*);
- At most 223 working days per year per employee (*Hard Constraint*);
- At most 22 Sundays/holidays worked per year per employee (*Hard Constraint*);
- Employee vacation periods must not be violated (*Hard Constraint*);
  The forbidden transitions are as follows:
  - Night → Morning | Night → Afternoon | Afternoon → Morning
- Each employee must be assigned only to allowed teams (*Hard Constraint*);

4 Cases Description

Number of Employees: 24

Case	Team Composition					
Case 1	A	10	B	10	AB	4

5 Problem Specification and Models

5.1 Minimum Staffing and Robustness

The problem under study consists of generating work schedules for a set of employees while satisfying operational requirements and legal constraints. The main objective is to ensure the required minimum and ideal staffing levels for each team and time interval (shift), while minimizing the number of unmet minimum and ideal staffing requirements throughout the planning horizon.

For each team, shift and time interval, a minimum and an ideal number of employees must be assigned. Violations of these staffing requirements are penalized in the objective function, whose overall goal is to minimize the total number of unmet requirements. Consequently, solutions with fewer unmet minimum staffing levels, and subsequently fewer unmet ideal staffing levels, are considered of higher quality.

5.2 Vacation Periods

In addition, the model explicitly considers predefined vacation periods. During these periods, employees cannot be assigned to shifts, reducing the overall workforce availability and increasing the combinatorial complexity of the problem. Incorporating vacation periods into the model is essential to ensure the practical feasibility of the generated schedules.

Vacation periods were generated sequentially for the 24 employees. Specifically, every 30 days, the next employee enters a vacation period, continuing this sequence throughout the year and restarting the cycle during the final month of the planning horizon.

5.3 Constraints

Besides staffing coverage and vacation periods, the model includes several structural constraints, described in Section 3. These constraints ensure that the generated schedules comply with both operational requirements and labor regulations.

5.4 Models

Three approaches are compared. The algorithms are described in the companion documents *(3 Shifts) Hybrid Puzzle Scheduling Heuristic* and *(3 Shifts) Greedy Scheduling Heuristics*.

- **Exact ILP.** A full-year Integer Linear Programming model that minimises the unmet minimum and ideal staffing requirements, solved with CBC.
- **Puzzle Heuristic.** A hybrid method that splits the year into periods of consecutive working days. In each period, an ILP chooses one work pattern and one team per employee from a catalogue of valid patterns, which can be limited by a pattern cap. Greedy phases then remove excess coverage, assign Sundays and holidays, and fix the remaining deficits.

- **Greedy heuristics (V1 and V2).** Constructive methods without solver. Both first cover the minimum requirements day by day, and differ in how they cover the ideal requirements: V1 is day-centric (it iterates over days and, for each day, over employees ranked by a score), while V2 is employee-centric (it iterates over employees and assigns each one to the pending day with the highest demand). A final pass assigns any remaining working days to employees who have not reached their annual maximum.

In summary, the problem combines staffing requirements, legal constraints, and individual employee availability limitations, resulting in a highly complex combinatorial optimization problem whose solution requires appropriate techniques to balance solution quality and computational efficiency.

6 Results

Throughout this report, η denotes the period length, in days, used by the Puzzle Heuristic.

The algorithm (ILP) produced the following optimal results using *CBC* solver in **24 seconds**:

Metric	**Value**
Minimums	103
Ideals	488

Table 1: Optimal results for 3 Shifts Problem.

Results Summary - Puzzle Heuristic

Solver	η **(days)**	**Time(s)**	**Minimums**	**Avg Ideals**	**Best Ideal**
CBC	2	3.8	103	585.02	559
	3	28.2	103	576.87	565
	4	33.7	103	574.6	556
	5	50.8	103	527.47	516
	6	82.2	103	573.4	553
	7	374.3	103	554.81	541
	8	504.4	103	578.98	562
	9	3534.9	103	547.89	533

Table 2: Performance of Puzzle Heuristic (without pattern cap)

Solver	η **(days)**	**Time(s)**	**Minimums**	**Avg Ideals**	**Best Ideal**	**Cap**
CBC	2	2.5	103	594.75	569	7
	3	6.8	103.02	581.19	569	26
	4	4	103.02	578.62	562	18
	5	53.9	103	525.65	512	201
	6	6.8	103	545.34	528	77
	7	23.5	103.09	515.11	502	139
	8	646.1	103.01	537.92	522	335
	9	155.9	103	520.08	506	694

Table 3: Performance of Puzzle Heuristic (with pattern cap)

Results Summary - Remaining Heuristics

Method	Time(s)	Minimums	Avg Ideals	Best Ideal
V1	0.30	103	558.57	536
V2	0.40	103	662.30	622

Table 4: Performance of the Greedy Heuristics (V1 and V2)

7 Comparative Analysis

This section compares the exact ILP model, the Puzzle Heuristic and the two Greedy variants (V1, V2) on the 24-employee, 3-shift instance. The *Minimums* metric is 103 for practically every method, so it does not discriminate between them. The comparison is therefore driven by the *Ideals* metric (lower is better), computational time, and stability across runs.

7.1 Solution Quality

Taking the exact optimum (488 ideal shortages) as reference, Table 5 reports the relative gap of the best and of the average solution found by representative configurations.

Method	**Best**	**Gap (best)**	**Avg**	**Gap (avg)**
ILP (exact, CBC)	488	0.0%	–	–
Puzzle, $\eta = 7$, with cap	502	2.9%	515.11	5.6%
Puzzle, $\eta = 9$, with cap	506	3.7%	520.08	6.6%
Puzzle, $\eta = 5$, no cap	516	5.7%	527.47	8.1%
Greedy V1	536	9.8%	558.57	14.5%
Greedy V2	622	27.5%	662.30	35.7%

Table 5: Relative gap to the exact optimum on the Ideals metric.

Two observations follow.

- **The exact model is the quality reference.** No heuristic reached 488, but the best Puzzle configurations come within about 3% of it, which is a strong result for a decomposition approach.
- **Puzzle versus Greedy.** With the cap, every configuration with $\eta \geq 5$ beats V1 on both the average and the best value. Without the cap, only $\eta = 5$, 7 and 9 beat the average of V1, and only $\eta = 5$ and 9 beat its best value. All Puzzle configurations beat V2 on both metrics. V2 is markedly worse (27.5% gap): its employee-centric strategy, which pops each employee's highest-demand day without a score-based ranking of who should fill it, is myopic and produces poorer global balance than the day-centric, score-driven V1.

Figures 1 and 2 show the average and the best ideal value of the Puzzle Heuristic as a function of the period length η. Since a longer period lets the ILP optimise more days jointly, a progressive decrease of the ideal value with η would be expected. The figures show that this is not what happens.

Without pattern cap (Figure 1), the average stays between 574.60 and 585.02 for $\eta = 2$ to $\eta = 4$ and then drops sharply at $\eta = 5$ (527.47, best 516), the lowest point of the curve and 8.1% above the optimum. Beyond that point the trend is lost: the average oscillates (573.40 at $\eta = 6$, 554.81 at $\eta = 7$, 578.98 at $\eta = 8$, 547.89 at $\eta = 9$), and the largest period tested ($\eta = 9$), which has the largest catalogue and the longest running time (3534.9 s), is still about 20 points worse than $\eta = 5$. The best-value curve follows the same pattern.

This suggests that, without a cap, the larger catalogues are not fully exploited by the solver within its stopping criteria.

With the pattern cap (Figure 2), the behaviour is more favourable but not monotonic either. The values for $\eta = 2$ to $\eta = 4$ are close to each other (594.75 to 578.62), followed by a sharp drop at $\eta = 5$ (525.65) and by an oscillation whose minimum is at $\eta = 7$ (515.11, best 502, only 2.9% above the optimum). The period $\eta = 9$ (520.08) is not better than $\eta = 7$, even though its catalogue is much larger (cap of 694 against 139). Unlike the previous case, every average value for $\eta \geq 5$ (at most 545.34) is lower than every value for $\eta \leq 4$ (at least 578.62). The effect of the cap is also uneven: it is large for $\eta = 6$ to $\eta = 9$ and negligible at $\eta = 5$ (527.47 versus 525.65).

Both figures show the same alternation, with valleys at $\eta = 5$, $\eta = 7$ and $\eta = 9$ and peaks at $\eta = 6$ and $\eta = 8$. Since it appears with and without cap, it is related to the period length itself and not to the cap. A plausible explanation is the interaction between the period boundaries and the structural constraints ($C_{\max} = 5$ consecutive days and the 7-day week, with Sundays and holidays handled separately), but this hypothesis was not tested. Only the average and the best value were recorded, so it is not possible to tell whether small differences between neighbouring values (for example, $\eta = 7$ and $\eta = 9$ with cap) are statistically significant.

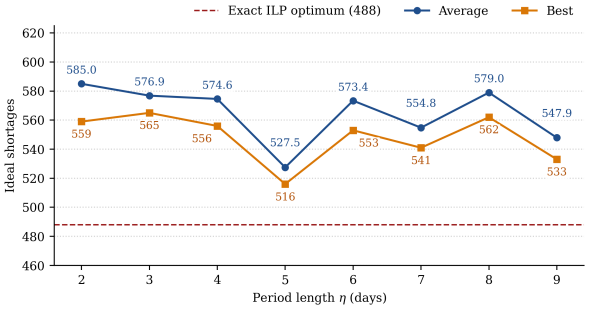


Figure 1: Average and best ideal value of the Puzzle Heuristic without pattern cap

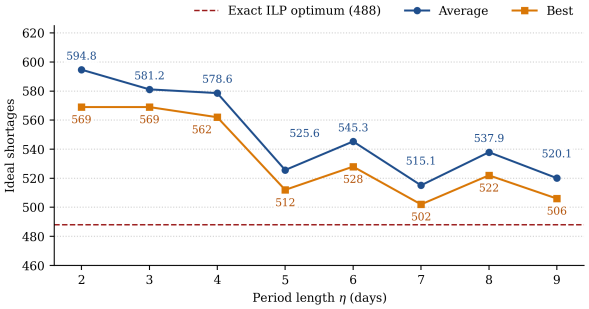


Figure 2: Average and best ideal value of the Puzzle Heuristic with pattern cap

7.2 Computational Efficiency

- **Growth of the Puzzle ILP.** Without a cap, the running time grows steeply with η, from 3.8 s ($\eta = 2$) to 374.3 s ($\eta = 7$) and 3534.9 s ($\eta = 9$). This is consistent with a rapid growth of the number of admissible patterns per employee with the period length.
- **Effect of the pattern cap.** The cap reduces the running time for most period lengths, and the reduction generally grows with η: from about 1.5× at $\eta = 2$ and 4× at $\eta = 3$, to 8× at $\eta = 4$, 12× at $\eta = 6$, 16× at $\eta = 7$ and 23× at $\eta = 9$ (3534.9 s to 155.9 s). This is consistent with the rapid growth of the catalogue: the larger it is, the more patterns the cap removes. Two cases depart from this trend. For $\eta = 5$ the time is essentially unchanged (50.8 s versus 53.9 s), which is compatible with a cap (201) that discards few of the admissible patterns. For $\eta = 8$ the time increased by about 28% (504.4 s to 646.1 s), even though the catalogue was reduced and the solution quality improved. A smaller catalogue therefore does not necessarily imply a faster solve. The causes of this behaviour were not investigated.

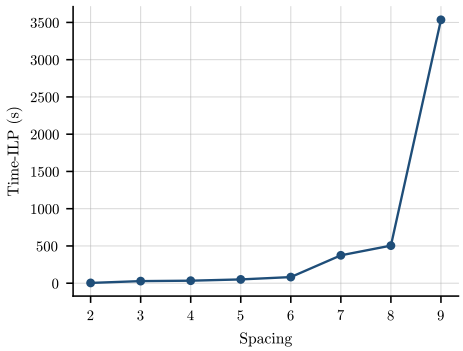


Figure 3: Running time of the Puzzle Heuristic without pattern cap

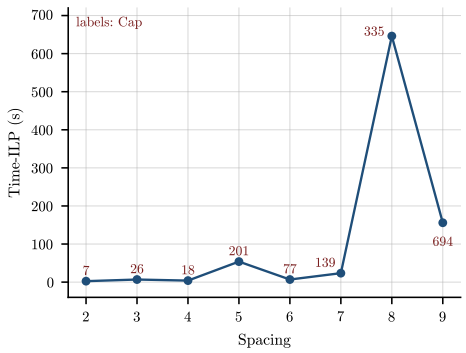


Figure 4: Running time of the Puzzle Heuristic with pattern cap

- **Cap improves quality for long periods.** For $\eta \geq 6$ the cap improves the average ideal value (e.g. $\eta = 9$: 547.89 → 520.08; $\eta = 7$: 554.81 → 515.11) and, except at $\eta = 8$, also saves time. At $\eta = 5$ the effect on quality is marginal (527.47 → 525.65). A possible explanation is that ranking patterns by proximity to the target τ removes unpromising candidates, so the solver, which stops at a 2% gap or at the time limit, spends its effort on a better region of the search space. For $\eta \leq 4$ the cap is slightly detrimental (e.g. $\eta = 2$: 585.02 → 594.75), possibly because the catalogue is already small and the cap discards useful patterns.

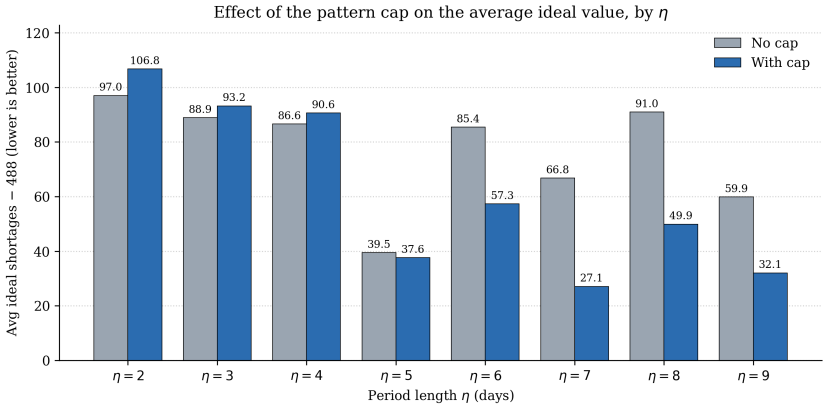


Figure 5: Relation between cap and average ideal over η (base line is the optimal value)

- **Greedy heuristics are very cheap per attempt.** The times reported for V1 and V2 are about 0.30 s and 0.40 s per attempt. This is about 80× (V1) and 60× (V2) faster than the exact ILP (24 s), and faster than every Puzzle configuration (2.5 s to 3534.9 s).
- **On this instance the exact ILP is competitive in time.** The full-year ILP was solved to optimality in 24 s, similar to the best Puzzle configuration ($\eta = 7$, cap: 23.5 s), and it yields a better solution.

Note on the measurement basis. The statistics of Minimums and Ideals (average and best) were computed over 3000 attempts of the greedy phases of the Puzzle Heuristic and over 50000 attempts of the Greedy heuristics. For the Puzzle Heuristic, the ILP stage runs once per configuration. Because only the greedy phases are repeated, additional Puzzle attempts are cheap, while each attempt of V1 and V2 is a complete run.

7.3 Consistency and Robustness

The gap between the average and the best solution measures how dependent a method is on the random components (tie-breaking, random day order). For the Puzzle Heuristic the gap between average and best ranges from about 11 to 26 points, for V1 it is 22.6 and for V2 it is 40.3. The following points are worth noting.

- **Best-of-many vs. single run.** The best values require many attempts (50000 for the Greedy and 3000 for the greedy phases of Puzzle heuristics). A user running a single attempt should expect the average, and there the Puzzle Heuristic ($\eta = 7$, cap) is about 8% better than V1 and 22% better than V2.
- **Feasibility on the minimums.** The Minimums metric is essentially constant (103), indicating that this value appears to be dictated by the instance itself (workforce capacity, vacations and hard constraints) rather than by the algorithm. Small deviations such as 103.02 or 103.09 with the cap show that, in a fraction of attempts, the reduced catalogue and the solver gap prevents the ILP + greedy phases from reaching the minimum coverage. This is a small but real robustness cost of the cap.
- **Feasibility on the minimums.** The Minimums metric is essentially constant (103), indicating that this value appears to be dictated by the instance itself (workforce capacity, vacations and hard constraints) rather than by the algorithm. Small

deviations such as 103.02 or 103.09 appear only with the cap, since without it every configuration averages exactly 103. A possible explanation is that, with the reduced catalogue, the ILP solution leaves less room for the randomised greedy phase that assigns Sundays and holidays, which in a fraction of the attempts ends with slightly more minimum shortages than the reference. This is a small but real robustness cost of the cap.

- **Constraint compliance.** All methods enforce the same hard constraints, so the differences are in soft-constraint quality only.

7.4 Scalability

- **Exact ILP.** The number of variables grows with employees, days, shifts and teams, and the combinatorial structure (consecutive days, shift transitions) makes the difficulty grow quickly with the instance size. For 24 employees it is solved in 24 s, but this may not hold for substantially larger workforces, more teams or more shifts, where memory and time can become prohibitive.
- **Puzzle Heuristic.** It converts one large problem into $|W|$ small ones, so the cost is controlled by η and by the pattern cap, not by the full horizon. Its bottleneck is the rapid growth of patterns with η, which is why the cap becomes essential for $\eta \geq 6$. The parameter η works as a tunable knob between speed and quality.
- **Greedy V1/V2.** Their complexity is polynomial in the number of employees and days, no solver is involved, and the measured cost is about 0.3–0.4 s per attempt on this instance. They are the natural choice for very large instances or when a solution is needed almost instantly, at the cost of quality.

7.5 Trade-offs

Criterion	**ILP**	**Puzzle**	**Greedy V1 / V2**
Solution quality	Optimal	Near-optimal (∼3–8%)	Moderate/poor (10–36%)
Time on this instance (per attempt)	24 s	2.5 s – 1 h (config.)	∼0.3–0.4 s
Variability	None	Moderate	Moderate/high
Scalability	Limited	Tunable	Very good
Tuning effort	Low	High (η, cap)	Low

Table 6: Qualitative summary of the trade-offs between the approaches.

8 Conclusions

On this 24-employee instance, the **exact ILP** gives the best solution (488 ideal shortages, within a 1% gap tolerance) in 24 s. No heuristic improves on it in quality, and only the Greedy variants and several Puzzle configurations (with the cap, $\eta = 2$, 3, 4, 6 and 7; without the cap, only $\eta = 2$) are faster, but with worse quality, so the heuristics offer no clear advantage here. Their value is expected to lie in larger or more constrained instances, where the exact model may become intractable; this was not tested in this work.

Among the heuristics, the **Puzzle Heuristic** gives the best solutions. The pattern cap is critical for $\eta \geq 6$: it reduces the running time by up to 23× (at η=9), except at η=8, where it increases it, and improves the average ideal value, while for $\eta \leq 4$ it slightly worsens it. The quality does not improve monotonically with the period length η. The best compromise is $\eta = 7$ with cap, with a best ideal value 2.9% above the exact solution (average of 515.11) and 23.5 s of running time. The cap values were tuned on the same instance on which the results are evaluated, so these results are in-sample, and the cap selection should be repeated for other instances.

Among the greedy methods, **V1** is a reasonably fast baseline (about 0.3 s per attempt) with moderate quality (best value 9.8% above the exact solution). **V2** is not competitive with V1: it takes more time and gives worse results (27.5% gap).

9 Limitations and Future Work

- **Single case.** Results come from one instance (24 employees, teams A/B/AB, one vacation pattern). Conclusions about scalability, and the expected advantage of the heuristics on larger instances, should be validated with more cases and larger workforces.
- **In-sample cap selection.** The cap values were tuned on the same instance on which the Puzzle Heuristic is evaluated. Their performance on different instances was not assessed.
- **Variability.** The ILP stage of each Puzzle configuration was run once, so the 3000 attempts capture only the variability of the greedy phases; that of the ILP stage (e.g. multithreading, time limit) was not measured. Only the average and best values were recorded, so standard deviations and significance tests between configurations (e.g. $\eta = 7$ and $\eta = 9$ with cap) were not possible.
- **Cap sensitivity.** Explore the sensitivity to the cap value and to η (in particular the non-monotonic behaviour observed), and test cap criteria other than proximity to τ.

List of Acronyms

ILP Integer Linear Programming. 1, 6

Glossary

CBC Open-source linear optimization solver specializing in Mixed-Integer Linear Programming (MILP). It is the standard computational engine for several modeling libraries due to its robustness and lack of licensing costs — Freeware. 1